Emiliano Mora Carrera, PhD

Computational Biologist | Bioinformatician | Data Scientist

Nationality: Mexican
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Availability: from August-September 2026

Education

PhD Evolutionary Biology (2018-2023)
University of Zurich
Zurich, Switzerland

MSc Biological Science (2014-2016)
National Autonomous University of Mexico
Mexico City, Mexico

BSc Biology (2009-2014)
University of Sonora
Hermosillo, Mexico

Technical Skills

Bioinformatic skills
Proficient: R, Bash (Shell), slurm, conda
Intermediate: git, Python, cloud computing
Basic: scikit-learn, Snakemake, Docker, SQL, Jupyter

Languages
Spanish (native)
English (professional proficiency)
German (A2)

Personal skills
Communication and presentation, advanced data visualization, mentoring, independent and collaborative work, adaptability

Professional summary

Computational biologist with 6+ years of experience integrating molecular biology, genomics, and bioinformatics to solve complex biological problems. Skilled in both wet-lab techniques (DNA extraction, PCR amplification, library preparation, sequencing) and the analysis of large-scale sequencing datasets, including short- and long-read data (Illumina and Nanopore) RNA-seq, Hi-C, and SSR markers. Experienced in developing reproducible and scalable workflows on HPC systems (slurm) using R, Bash, and conda. Passionate about data analysis, visualization, and applying computational methods to answer biological questions. Committed to scientific communication, collaborative work and open science.

Professional Experience

Postdoctoral Researcher – Computational Genomics
University of Zurich, 2023 to present

- Designed and implemented scalable bioinformatics pipelines for comparative genomic and transcriptomic analyses across 60+ plant genome assemblies, enabling large-scale evolutionary and functional genomics studies.
- Conducted genome-wide association studies (GWAS), population genomics, transcriptomic analyses, and phylogenetic inference in *Erysimum cheirathoides* and related species to identify evolutionary and genomic patterns across populations.

- Generated a chromosome-scale reference genome for the tetraploid species *Primula grandis* by integrating short-read, long-read, and Hi-C sequencing technologies; additionally performed tissue-specific transcriptomic analyses to characterize subgenome-level expression differences.
- Processed and analyzed 100+ whole-genome sequencing datasets across 10 Primula species, integrating variant calling, sequence alignment, and evolutionary modeling to reconstruct the evolutionary history of *P. grandis*.
- Set up, optimized, and maintained high-performance computing (HPC) infrastructure at the University of Zurich to support large-scale genomics and bioinformatic analyses.

Doctoral Researcher – Evolutionary Genomics
University of Zurich

- Designed and implemented reproducible bioinformatic pipelines for population genomics analyses and phylogenetic inference across large-scale sequencing datasets.
- Processed and analyzed 100+ whole-genome sequencing datasets, integrating variant calling, sequence alignment, and evolutionary modeling workflows to investigate genomic diversity and evolutionary relationships.
- Collaborated with multidisciplinary international research teams, contributing to the publication of 10 peer-reviewed studies in high-impact journals.
- Applied statistical and computational models to analyze complex biological datasets, generating reproducible, scalable, and data-driven insights into genomic and evolutionary processes.

Research Assistant – Plant-Animal Interactions Lab
Institute of Ecology, UNAM

- Performed DNA extraction for > 2,000 samples. Amplified, optimized and analyzed microsatellites (ISSRs).

Selected Publications (link to Google Scholar)

- Mora-Carrera, et al. (2025). Genomic patterns of loss of distyly and polyploidization in primroses. **Molecular Biology and Evolution**.
- Mora-Carrera E, et al. (2024). *Unveiling the genome-wide consequences of range expansion and mating system transitions in Primula vulgaris.* **Genome Biology and Evolution.**
- Mora-Carrera E, et al. (2021). *Origins and consequences of independent shifts to homostyly within species.* **Molecular Ecology.**

Teaching & Mentorship

- Designed and delivered lectures in *Pollination Ecology* (UNAM, Mexico).
- Served as Teaching Assistant in multiple courses on *Molecular Evolution, Ecological Genomics*, and *Genome Evolution* (University of Zurich, 2018–2024).

- Developed hands-on tutorials for bioinformatics methods. Guided and mentored undergraduate and master’s students in computational biology projects in bioinformatic platforms, supporting them in data analysis, coding, and reproducible workflows.

Grants & Awards

- Swiss Government Excellence Scholarship (2018–2021) – 70,000 CHF
- Forschungskredit Candoc UZH (2021) – 43,160 CHF

Volunteering

- Pint of Science (2019 and 2022)
- Zurich Film Festival (2025)
- Socialwerk Pfarrer Sieber (2025-2026)